Assignment 2

Total = 10 Marks

Multiple Choice Questions

Question 1:

When a sample sapae is partitioned into n mutually exclusive and exhaustive events $A_1, A_2, \ldots A_n$, the law of total probability of an event B is given by

(a) $P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)$
(b) $P(B) = \sum_{i=1}^{n} P(B \cap A_i)$
(c) $P(B) = \sum_{i=1}^{n} P(B|A_i)$
(d) $P(B) = P(A_i)P(B)$

[1 *Mark*]

Correct Answer: (a) $\boldsymbol{P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)}$

Detailed solution:

Based on the theorem of total probability,

$$\boldsymbol{P(B) = \sum_{i=1}^{n} P(B|A_i)P(A_i)}$$

Question 2:

Three events E_1, E_2, and E_3 are mutually exclusive and collectively exhaustive. The corresponding probabilities are as follows, $P(E_1) = 0.3$, $P(E_2) = 0.4$, then $P(E_3)$ is:

(a) 0.1
(b) 0.2
(c) 0.3
(d) 0.4

[1 *Mark*]

Correct answer: (c) 0.3

Detailed solution:

Given that the three events are mutually exclusive and collectively exhaustive events. Therefore, the sum of the three event probabilities must be equal to the entire sample space i.e.,

$$P(E_1) + P(E_2) + P(E_3) = 1$$

Therefore, $P(E_3) = 1 - P(E_1) - P(E_2) = 1 - 0.3 - 0.4 = 0.3$

Question 3:

A data scientist observes that 35% of machine learning models achieve an accuracy greater than 90%, while 40% of models achieve an F1-score greater than 0.90. It is also found that 25% of the models satisfy both criteria. What is the probability that a randomly selected model satisfies at least one of these two criteria?

(a) 0.75
(b) 0.11
(c) 0.50
(d) 0.90

[2 *Mark*]

Correct answer: (c) 0.50

Detailed solution:

Let A be the event of the data scientist observing the accuracy and F be the event of the data scientist observing the F1-score.

Given,

The probability that a data scientist observes an accuracy of 0.90 is, $P(A) = 0.35$.

The probability that a data scientist observes F1-score of 0.90 is, $P(F) = 0.40$.

The analysis shows that the probability of models satisfying both criteria is $P(A \cap F) = 0.25$.

Let $P(A \cup F)$ be the probability that a randomly selected model satisfies at least one of these performance criteria, then

$$P(A \cup F) = P(A) + P(F) - P(A \cap F) = 0.35 + 0.40 - 0.25 = 0.50$$

Therefore, the probability that a randomly selected model satisfies at least one of these performance criteria is $P(A \cup F) = 0.50$.

Question 4:

A crop insurance company observes that the probability of crop failure is 0.30 in a vulnerable region, whereas it is 0.10 in a non-vulnerable region. It is also known that 50% of the crops belong to the vulnerable region. If a randomly selected insured crop is found to have failed, what is the probability that it belongs to the vulnerable region?

(a) 0.55
(b) 0.65
(c) 0.75
(d) 0.85

[3 *Mark*]

Correct answer: (c) 0.75

Detailed solution:

Given 50% of the crop belongs to the vulnerable region, i.e., the probability of getting a crop in the vulnerable zone is, $P(V) = 0.5$. Therefore, another 50% crop belongs to the non-vulnerable regions, i.e., the probability of getting a crop in the non-vulnerable zone is, $P(V') = 0.5$.

The probability of crop failure in vulnerable region is, $P(F|V) = 0.3$.

Similarly, the probability of crop failure in non-vulnerable region is, $P(F|V') = 0.1$.

Let $P(V|F)$ be the probability of randomly selected insured crop is failed, given that it belongs to the vulnerable region,

$$P(V|F) = \frac{P(F|V) \times P(V)}{P(F|V) \times P(V) + P(F|V') \times P(V')} = \frac{0.3 \times 0.5}{0.3 \times 0.5 + 0.1 \times 0.5} = 0.75$$

Therefore, the probability of a randomly selected insured crop is failed, given that it belongs to the vulnerable region $P(F|V)$ is **0.75**.

Question 5:

A medical screening test correctly detects a disease with probability 0.92 among high-risk patients and 0.85 among low-risk patients. It is known that 40% of the patients belong to the high-risk group. If a randomly selected patient is correctly diagnosed, the probability that the patient belongs to the high-risk group is _____ .

[3 *Mark*]

Correct answer: Range: 0.40 to 0.43

(0.419)

Detailed solution:

Given, 40% of the patients belong to high risk i.e., the probability of getting a high-risk patient is $P(H) = 0.4$. Therefore, 60% of the patients belong to low risk i.e., the probability of getting a low-risk patient is $P(H') = 0.6$.

Let $P(D|H) = 0.92$ be the probability of detecting the disease among high-risk patients. Similarly, $P(D|H') = 0.85$ be the probability of detecting the disease among low-risk patients.

Let $P(H|D)$ be the probability of correctly detecting the disease given that the patient belongs to a high-risk group. Using Bayes theorem,

$$P(H|D) = \frac{P(D|H) \times P(H)}{P(D|H) \times P(H) + P(D|H') \times P(H')} = \frac{0.92 \times 0.4}{0.92 \times 0.4 + 0.85 \times 0.6} = 0.419$$

Therefore, the probability of correctly detecting the disease given that the patient belongs to a high-risk group $P(D|H)$ group is **0.419**.